

Connectionist School of CRUM

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I. CONNECTIONIST SCHOOL OF CRUM

CRUM stands for the Computational Representational understanding of the mind. This theory explains that the mind works just like a computer would, where it would take in information from mental states and this information would then be represented in the brain and then computations and operations would take place to make sense of things and solve problems [1]. CRUM includes different approaches that explain how the thinking and procedures work, such as logic and connectionism.

The connectionist school of CRUM explains that the way our mind works is through what is similarly known as neural networks. In a neural networks there a group of connections through units, in which case the connectionism explains to be neurons [2]. Our neurons work together like a neural network to send signals between one another. The information processed through these networks, help build connections through the activation of the neurons, so when you are learning something new or solving a problem, the connections between specific neurons strengthen. This adjusting of the connecting neurons and the weight of the connection between them causes the different experiences within our brain. This explains the thinking and learning through patterns from the connectionism school of CRUM, just like in artificial intelligence how neural networks learn to understand patterns and draw conclusions.

II. BRAIN SCHOOL

The Brain School, unlike CRUM, focuses on more of the physical and biological aspect of the mind to understand cognition. To explain cognition, the school emphasizes more of things like the cellular level of the brain and the actual biological components that goes into thought rather than the analogy of neural networks. Instead of treating the mind as a computer, it argues that we must study the neurons, brain, and biological processes. The school analyzes and studies how neurons actually fire within the brain, what chemical processes the brain goes through during certain emotions and states, and also how brain chemistry affects thoughts and behaviors. So, ultimately, CRUM focuses on computational models to explain how the mind operates, the brain school identifies the mind to be a product of physical, biological, and chemical processes that happens within the brain itself.

III. ARE BRAINS CONNECTIONIST?

Personally, I believe that the answer to the question of whether or not brains are connectionist cannot be answered

with a simple yes or no. However, if it needs to be either yes or no then the answer would be yes. The brain would most certainly be considered a connectionist mainly due to the nature of information its processed. The brain processes information through a neural network built of neurons that represent and transmit information [2]. Just like a neural network, the network of neurons fire off signals to one another at different weight and form different patterns and experiences. This all put together within the brain to be able to develop different thoughts and processes within the mind. This is undoubtedly the procedures that occurs within the brain during thinking and definitively supports that brains are connectionist.

However, the human brain is not just a connectionist if looked at as a whole. The connectionism does not explain more in how the brain works if it over looks the biological and physical aspects fo the brain. The brain operates with chemical and biological processes to support it, not just through a neural network of neurons alone. While the brain is indeed a connectionist, theories like the brain school also has supporting data that is true as well [3]. So the brain's complete understanding of how the mind works would need to be a combination of these theories.

IV. CONUNDRUM

The conundrum ultimately comes from the fact that although connectionism is a part of CRUM the brain school rejects CRUM itself. Since brains are connectionist, this shouldn't be logical. CRUM looks at the mind and posits that thought is computation from representations of mental states. The brain school on the other hand states that cognition is the result from biological processes within the brain. It shows that CRUM is only focused on the computational side of the mind without accounting for the physical [4]. If brains are connectionist then technically the brain school should not reject CRUM. However, looking at it another way is that connectionism is only a small part of CRUM itself. CRUM as a whole sees the mind as like a computer, where information is taken in and then represented to solve computational problems. Connectionism is only a part of the whole. Although the brain school rejects CRUM, it rejects the theory in that cognition solely stems from the abstract alone. The brain school recognizes that the physical aspect of the brain plays a big part of thought and cognition; however, the school understands that it is an overlap of abstract computations and biological processes that combined together creates thought, unlike CRUM that focuses on just computation. This fixes the

conundrum itself in which the brain school rejects CRUM as a theory overall; however, it understands and agrees with certain aspects of the model.

REFERENCES

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